SKILL BUILDERS · UNIT 1 · SCORE 3

Real Numbers: Practice Set

NC.8.NS.1–2 · NC.8.EE.2

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.NS.1

Is $0.1212\ldots$ rational or irrational?

(A) Rational (B) Irrational

2 Score 3 NC.8.NS.1

Is $\sqrt{144}$ rational or irrational?

(A) Rational (B) Irrational

3 Score 3 NC.8.NS.1

Is $\sqrt{8}$ rational or irrational?

(A) Irrational (B) Rational

4 Score 3 NC.8.NS.1

Write $0.\overline{7}$ as a fraction.

(A) 7/9 (B) 7/10

(C) 1/7 (D) 7/99

5 Score 3 NC.8.NS.1

Write $0.\overline{45}$ as a fraction in simplest form.

(A) 45/100 (B) 5/11

(C) 45/99 (D) 9/20

6 Score 3 NC.8.EE.2

Find $\sqrt{121}$.

Answer: _____

7 Score 3 NC.8.EE.2

Find $\sqrt[3]{64}$.

Answer: _____

8 Score 3 NC.8.NS.2

$\sqrt{40}$ is between which two whole numbers?

(A) 20 and 21 (B) 6 and 7

(C) 5 and 6 (D) 36 and 49

9 Score 3 NC.8.NS.2

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: _____

10 Score 3 NC.8.NS.2

Order from least to greatest:

$\pi,\ \sqrt{10},\ 10/3,\ 3.5$

(A) √10, π, 10/3, 3.5

(B) π, √10, 10/3, 3.5

(C) 3.5, 10/3, √10, π

(D) π, 10/3, √10, 3.5

11 Score 3 NC.8.EE.2

Find the mistake. Sam wrote: **√36 = 18**. What went wrong?

(A) Sam divided 36 by 2. √36 = 6 because 6 × 6 = 36.

(B) Sam is right: 18 + 18 = 36.

(C) √36 = 9 because 4 × 9 = 36.

(D) √36 = 72 because 36 × 2 = 72.